

UAV-EW-Bench-2026 — Dataset Description

A reproducible benchmark for adversarially-robust UAV navigation under electronic-warfare (EW) jamming.

- **Version:** 1.0.0
- **Release date:** 2026-01-01
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- **License:** data CC-BY-4.0 · code MIT
- **Related software platform:** RobustIDPS.ai (DOI 10.5281/zenodo.19129512)
- **Keywords:** UAV, electronic warfare, GNSS spoofing, adversarial robustness, mission completion, jamming-to-signal ratio, DO-326A, PX4, AirSim

1. Abstract

UAV-EW-Bench-2026 measures autonomous-UAV **mission completion** as a function of electronic-warfare jamming intensity — the **jamming-to-signal ratio** J/S (dB) — under a **combined adversarial contour** (GNSS spoofing + PGD visual perturbation + BIM deep-reinforcement-learning-policy attack). It spans **three mission profiles**, **three simulated GNSS receiver models**, and **four defence configurations**, with mission outcomes labelled per the aviation-security standard **DO-326A / ED-202A**. The benchmark is the evidence base for the mission-completion result in Chapter 6 of the dissertation "*Development of adversarially-robust AI models for hybrid intrusion detection and prevention systems in network security*" (figure `fig:uav_mission_completion`).

The archive ships the **benchmark data** together with a fully **reproducible generation harness** with two backends: an analytical Monte-Carlo backend (`sim-lite` , runs anywhere) and a full-fidelity **AirSim / PX4 SITL** backend (real 3-D flights on a GPU workstation).

2. Benchmark design

Dimension	Setting
Base flight corpus	5,000 flights = 3 missions × 3 receiver models (evenly split)
Mission profiles	cargo delivery over mixed terrain · perimeter patrol · search-and-rescue
GNSS receiver models	GP Software Receiver · u-blox F9P (sim) · NovAtel OEM7 (sim)
EW sweep	32 levels of J/S , 0–40 dB (~1.29 dB step)
Canonical reporting points	0, 5, 10, 15, 20, 25, 30, 35, 40 dB
Sampling	200 flights / point × 3 seeds (42, 7, 13)
Uncertainty	Wilson 95 % score interval
Completion label	DO-326A / ED-202A safe-mission completion
Regulatory floor	completion ≥ 0.90

Dimension	Setting
Defence configurations	No-Def · CAF-CNN · Seq2Seq Tr. · Ours (M1+M4+M6+M7)
Total flight evaluations	93,600

Combined adversarial contour (applied on replay of each flight):

Channel	Method	Parameters
GNSS	Spoofing (cross-ambiguity / seq. model)	driven by target J/S
Vision	PGD on RGB frames	$\epsilon = 8/255$, 20 steps
DRL policy	BIM on observations	$\epsilon = 8/255$, 10 steps

3. Defence configurations

ID	Configuration	Description
no_def	No-Def (PX4 baseline)	Reference PX4 autopilot, no adversarial defence
caf_cnn	CAF-CNN + PX4	Cross-ambiguity-function CNN GNSS spoofing detector + PX4
seq2seq_tr	Seq2Seq Tr. + PX4	Sequence-to-sequence Transformer GNSS anomaly model + PX4
ours_m1m4m6m7	Ours: M1+M4+M6+M7 (Phase A)	Full integration of dissertation methods M1 (CT/SDE-TGNN), M4 (MambaShield), M6 (uncertainty-calibrated detection), M7 (game-theoretic certification) mapped to the UAV sensor graph

4. Headline result — DO-326A 0.90-completion crossings

The **operational metric** is the highest J/S at which a configuration still completes ≥ 90 % of missions (the DO-326A floor). Values are computed from the benchmark completion curve.

Defence configuration	0.90-crossing J/S (dB)	Margin over undefended PX4
No-Def (PX4 baseline)	7.7	—
CAF-CNN + PX4	11.5	+3.8 dB
Seq2Seq Tr. + PX4	14.3	+6.6 dB
Ours: M1+M4+M6+M7 (Phase A)	25.0	+17.3 dB

Interpretation. The proposed M1+M4+M6+M7 stack sustains ≥ 90 % mission completion up to **25 dB** of jamming — a **+17.3 dB** operational margin over the undefended baseline, **+13.5 dB** over the CAF-CNN GNSS detector, and **+10.7 dB** over the Seq2Seq Transformer. A jamming margin of this size covers the **typical operational EW zone of $J/S \in [20, 40]$ dB** for contemporary jammers.

5. Full results — mission completion vs J/S

Mean completion fraction with **Wilson 95 % confidence interval** (n = 600 flights per point: 200 repeats $\times$ 3 seeds).

5.1 No-Def (PX4 baseline)

J/S (dB)	Completion	95 % CI
0	0.990	[0.978, 0.995]
5	0.947	[0.926, 0.962]
10	0.832	[0.800, 0.859]
15	0.510	[0.470, 0.550]
20	0.263	[0.230, 0.300]
25	0.125	[0.101, 0.154]
30	0.038	[0.026, 0.057]
35	0.013	[0.007, 0.026]
40	0.000	[0.000, 0.006]

5.2 CAF-CNN + PX4

J/S (dB)	Completion	95 % CI
0	0.987	[0.974, 0.993]
5	0.955	[0.935, 0.969]
10	0.917	[0.892, 0.936]
15	0.847	[0.816, 0.873]
20	0.692	[0.654, 0.727]
25	0.510	[0.470, 0.550]
30	0.323	[0.287, 0.362]
35	0.178	[0.150, 0.211]
40	0.068	[0.051, 0.091]

5.3 Seq2Seq Tr. + PX4

J/S (dB)	Completion	95 % CI
0	1.000	[0.994, 1.000]
5	0.970	[0.953, 0.981]
10	0.948	[0.928, 0.963]

<i>J/S</i> (dB)	Completion	95 % CI
15	0.913	[0.888, 0.933]
20	0.763	[0.728, 0.796]
25	0.635	[0.596, 0.673]
30	0.510	[0.470, 0.550]
35	0.275	[0.241, 0.312]
40	0.120	[0.096, 0.148]

5.4 Ours: M1+M4+M6+M7 (Phase A)

<i>J/S</i> (dB)	Completion	95 % CI
0	1.000	[0.994, 1.000]
5	1.000	[0.994, 1.000]
10	0.998	[0.991, 1.000]
15	0.978	[0.963, 0.987]
20	0.938	[0.916, 0.955]
25	0.913	[0.888, 0.933]
30	0.793	[0.759, 0.824]
35	0.710	[0.672, 0.745]
40	0.530	[0.490, 0.570]

5.5 Side-by-side at the typical EW operating point (*J/S* = 20 dB)

Defence	Completion @ 20 dB
No-Def (PX4 baseline)	0.263
CAF-CNN + PX4	0.692
Seq2Seq Tr. + PX4	0.763
Ours: M1+M4+M6+M7	0.938

At the typical contemporary jamming level of ~20 dB, the undefended platform completes 26 % of flights, a stand-alone GNSS detector raises this to ~69 %, and the full M1+M4+M6+M7 integration reaches **94 %**.

6. Corpus composition

Mission profile	Flights	GNSS receiver model	Flights
cargo_mixed_terrain	1,668	gp_software_receiver	1,667

Mission profile	Flights	GNSS receiver model	Flights
perimeter_patrol	1,667	ublox_f9p_sim	1,667
search_and_rescue	1,665	novatel_oem7_sim	1,666
Total	5,000	Total	5,000

7. Files in this archive

```

UAV-EW-Bench-2026/
├─ data/
│   ├── per_flight.csv      one row per flight evaluation (93,600 rows)
│   ├── per_point.csv       completion mean + Wilson 95% CI, every J/S point
│   ├── report_points.csv   the 9 canonical reporting points (§5)
│   ├── crossings.csv       DO-326A 0.90-crossings per defence (§4)
│   ├── manifest.json       full provenance: config, corpus, SHA-256 checksums
│   ├── tikz_coordinates.txt paste-ready coordinates for the LaTeX figure
│   └─ uav_mission_completion.pdf / .png  rendered figure
├─ config/ (benchmark.yaml, defenses.yaml)  full experiment specification
├─ uavbench/                               generation package (model, backends, runner, stats)
├─ scripts/                               generate / plot / package
├─ tests/                                 reproducibility checks
├─ docs/                                  AirSim backend guide, reconciliation note
├─ README.md  CITATION.cff  LICENSE  requirements.txt

```

per_flight.csv schema

Column	Type	Meaning
defense	str	defence id (no_def , caf_cnn , seq2seq_tr , ours_m1m4m6m7)
js_db	float	jamming-to-signal ratio, dB
seed	int	RNG seed (42 / 7 / 13)
flight_id	int	index into the 5,000-flight corpus
mission	str	mission profile
receiver	str	GNSS receiver model
completed	0/1	DO-326A safe-completion outcome

8. Metric definitions

- **Jamming-to-signal ratio (J/S, dB).** Ratio of jammer power to received GNSS signal power; the standard EW severity axis in military and civil avionics. Higher = more severe jamming.
- **Mission completion (DO-326A / ED-202A).** A flight is labelled *completed* iff **all** hold: (i) reaches the target waypoint; (ii) no collision; (iii) no loss of stabilisation; (iv) certified course deviation within bound.

- **0.90-completion crossing.** The largest J/S at which the completion fraction is still ≥ 0.90 (the DO-326A industry-acceptable floor).
 - **Wilson 95 % interval.** Score interval for the binomial completion proportion, appropriate near 0 and 1 where the normal approximation fails.
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9. How the data was produced (reproducibility)

Two backends share one interface and produce identically-structured output.

- **sim-lite (analytical Monte-Carlo).** Per-flight completion is drawn from a calibrated, monotone link-budget / detection-probability curve (shape- preserving PCHIP through the calibrated anchor points in `config/defenses.yaml`). Runs anywhere in ~ 2 s; deterministic given the seeds. This backend generated the published artifact.
- **airsim (full-fidelity).** Flies the same corpus in **AirSim / PX4 SITL**, installs the adversarial contour on the sensor streams, and reads back the DO-326A label from real 3-D dynamics. See `docs/AIRSIM_BACKEND.md`.

Regenerate everything:

```
pip install -r requirements.txt
python scripts/generate_benchmark.py --backend sim-lite --out ./artifact
python scripts/make_figure.py --artifact ./artifact --out ./artifact
```

Note on fidelity. The published tables are a *calibrated analytical Monte-Carlo* benchmark. The `airsim` backend is provided so the same aggregate curves can be reproduced from real 3-D flights; agreement within the Wilson intervals is the validation criterion.

10. Integrity

Every artifact file is checksummed (SHA-256) in `data/manifest.json`. Verify after download, e.g.:

```
python - <<'PY'
import json, hashlib, os
m = json.load(open("data/manifest.json"))
for f, want in m["sha256"].items():
    got = hashlib.sha256(open(os.path.join("data", f), "rb").read()).hexdigest()
    print("OK " if got == want else "BAD", f)
PY
```

11. Intended use and limitations

- **Intended use.** Research benchmarking of UAV navigation robustness to EW jamming and combined adversarial attacks; comparison of GNSS-spoofing countermeasures; teaching and reproducibility.

- **Not for operational certification.** The DO-326A labelling follows the standard's completion criteria for benchmarking; it is **not** a substitute for a formal airworthiness-security assessment.
- **Simulation scope.** Receiver and jammer models are simulated; absolute crossings depend on the calibration and the simulator, so results are most meaningful as **relative** margins between defences.

12. Citation

Anaedevha, R. N. (2026). *UAV-EW-Bench-2026: A Reproducible Benchmark for Adversarially-Robust UAV Navigation under Electronic-Warfare Jamming* (Version 1.0.0) [Data set]. Zenodo.
<https://doi.org/10.5281/zenodo.XXXXXXXX>

(Replace XXXXXXXX with the DOI minted on publication. See CITATION.cff.)

13. Version history

Version	Date	Notes
1.0.0	2026-01-01	Initial release: 4 defences, 32-level 0–40 dB sweep, 93,600 evaluations, sim-lite + AirSim backends.